

Fun Time Website

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Abstract

Fun Time Website

The “Fun Time Website” is an engaging and user-friendly platform designed to offer a wide range of adventure and recreational activities for different user groups, including Kids, Adults, and adrenaline-seeking Fun Players. Users can explore and book activities such as skydiving, hiking, tree climbing, paragliding, and more, tailored to their age and interests. The website provides a seamless experience, allowing users to easily search for events, view detailed information, and securely book their chosen activities. Admins can efficiently manage events and monitor user interactions, ensuring a smooth and enjoyable experience for all users.

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Chapter 1

Requirements

1.1 Non-Functional Requirements:

- Performance: Use cases like (Search and Filtering) and (View Event Details) involve actions where response time is crucial. By identifying these actions, use-case modeling helps ensure the system can handle these tasks quickly, guiding optimizations for faster performance.

- Security: In use cases such as (Booking and Purchasing), where payment and personal data are handled, use-case modeling emphasizes where security protocols like encryption and secure authentication are necessary to protect user information.

- Availability: Use-case modeling shows critical functionalities, such as (Admin Management) and (Booking and Purchasing), that require high availability. This helps developers plan for system stability to minimize downtime, especially for time-sensitive activities.

- Accessibility: Use cases illustrate each step in the user's journey, supporting designs that are intuitive and adaptable across devices and user needs, ensuring a more inclusive experience.

- Maintainability: By breaking down each feature into clear use cases, use-case modeling helps create a modular system structure. This ensures code can be easily updated, making it more manageable for future enhancements and fixes.

1.2 Functional Requirements:

- The system implements a robust registration and login feature that validates user credentials, provides error feedback, and achieves a 98% success rate in secure account access.

- The system develops an account management system with encrypted password storage, secure login, password reset functionality, and account deactivation, maintaining zero vulnerabilities to security threats.

- The admin creates a responsive search and filter feature allowing users to filter activities by age group and personal preferences, with search results loading within 2 seconds.

- The system processes bookings and handles payment transactions securely, achieving a 99% completion rate without errors.

- The system enables users to set and update preferences and profile information, with changes reflected immediately.

- The system implements a reliable notification system that sends reminders for upcoming booked activities, with 95% of messages delivered at least 24 hours in advance.

- The admin uploads and manages activity descriptions and images. The system provides comprehensive activity details with descriptions and images that load in under 2 seconds, ensuring users have access to essential information.

- The system collects and displays user ratings and reviews instantly, maintaining a 1-second load time. The admin moderates and manages reviews to ensure compliance with platform guidelines.

1.3 Stake Holders:

Third parties: Payment services(Paypal, master card,etc).
Event providers.

1.4 Context Diagram

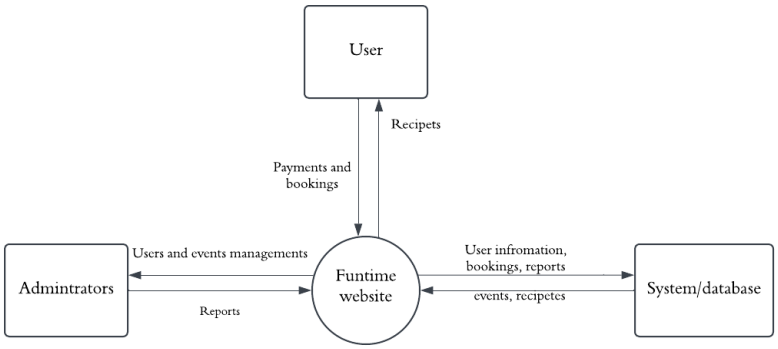


Figure 1.1: Context Diagram

1.5 ER Diagram

1.6 Class Diagram

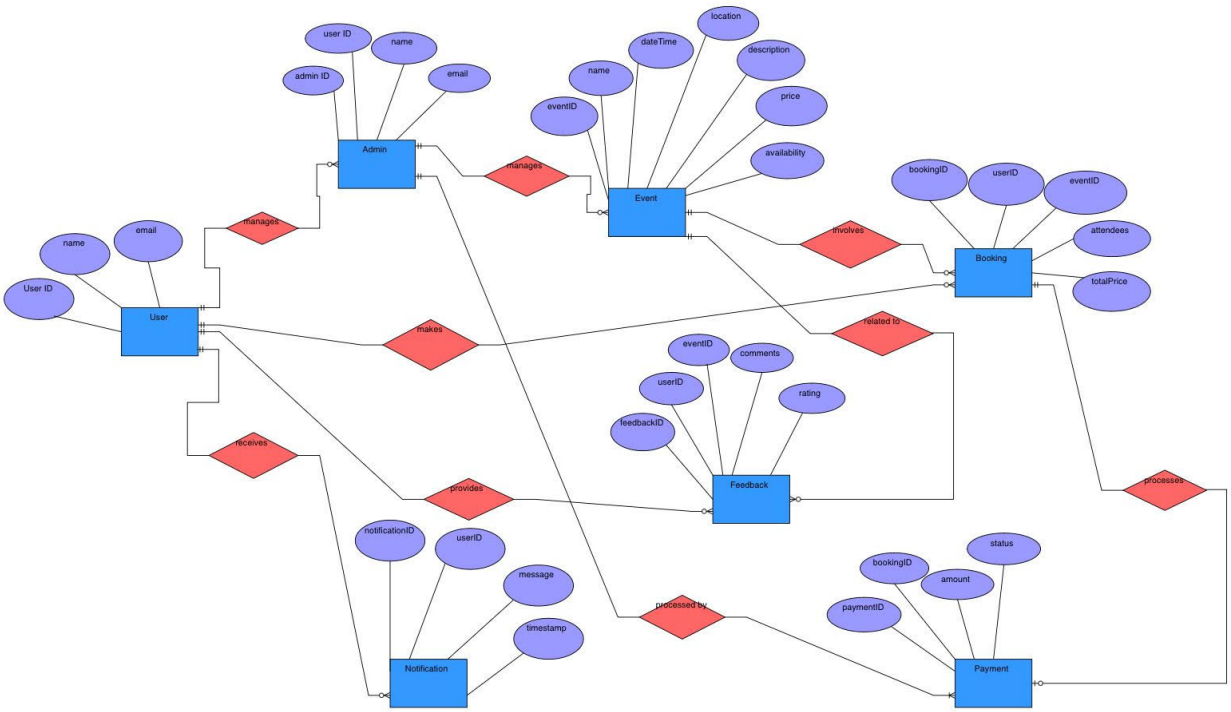


Figure 1.2: ER Diagram

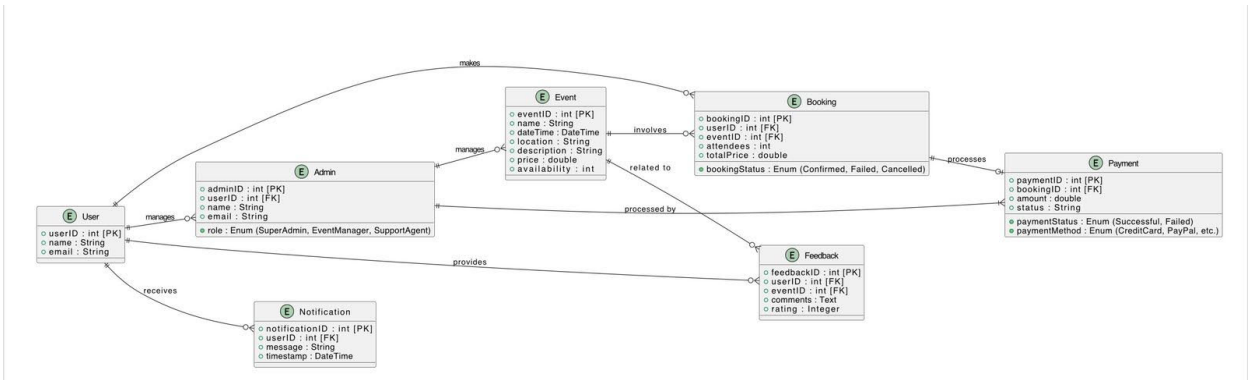


Figure 1.3: ER Diagram 2

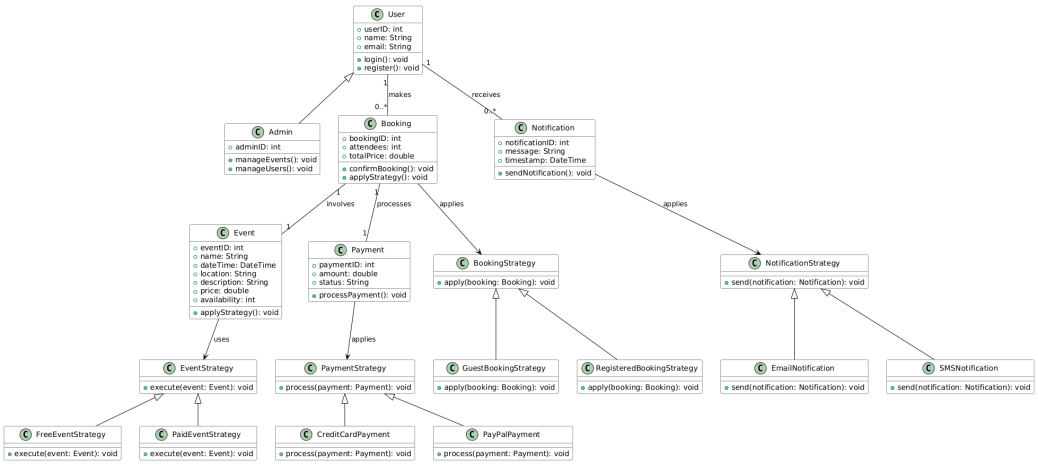


Figure 1.4: Class Diagram

Chapter 2

Traceability Matrix

Requirements	PW	UC-1: User Authentication and Management	UC-2: Search and Filtering	UC-3: Booking and purchasing	UC-4: View Event Details	UC-5: Admin Management
REQ1: User registration and login functionality	5	X				
REQ2: Secure user authentication and account management	4	X				
REQ3: Search and filter activities based on age group and preferences	4		X			
REQ4: Activity booking and payment processing	5			X		
REQ5: Managing user preferences and profiles	3	X				
REQ6: Sending notifications and reminders for booked activities	3					X
REQ7: Viewing activity details and descriptions	2		X		X	
REQ8: Collecting and displaying user ratings and reviews	1					X
Max PW		5	4	5	3	3
Total PW		12	8	5	5	4

Figure 2.1: Traceability Matrix

Chapter 3

Use cases

3.1 Use case 1: User Authentication and Management

Related Requirements:

1. Secure login and account management
2. User registration with validation checks

Initiating Actor:
User (Adults, Fun Players, Parents for kids)

Actor's Goal:
To log in or create a new account for managing personal settings.

Participating Actors:
Administrators
Preconditions:
The user has an existing account or is creating a new one.

Post-conditions:
The user is successfully logged in and can access the account dashboard.

Constraints:
-Enforce secure password storage and strong password policies.
-Limit login attempts to prevent unauthorized access.

Flow of Events for Main Success Scenario:

1.The user navigates to the login page
2.The user enters login credentials (username and password) or selects 'Create Account' for new registration.
3.The system validates login information:
- If credentials are correct, the user is directed to their dashboard.
- If 'Create Account' is selected, the user is prompted for registration details.
4.The system securely stores user details and grants access to the dashboard upon successful registration.

Flow of Events for Extensions (Alternate scenarios):

1a. Login fails due to incorrect credentials; the system displays an error, allowing the user to retry or recover the password.
2a. Registration info does not meet security requirements; the system prompts for corrections.

Use case diagram:

Use Case Sequence:

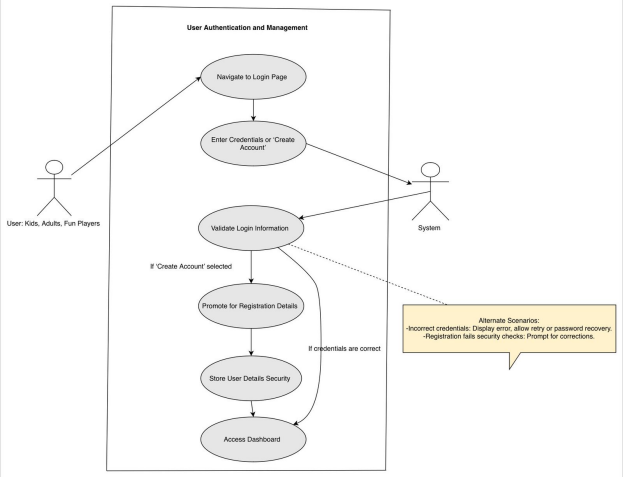


Figure 3.1: UC-1:User authentication and management

Use Case State:

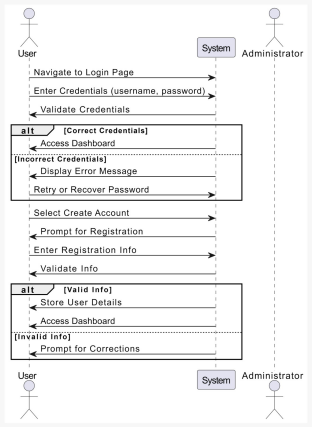


Figure 3.2: UC1:User authentication and management sequence diagram

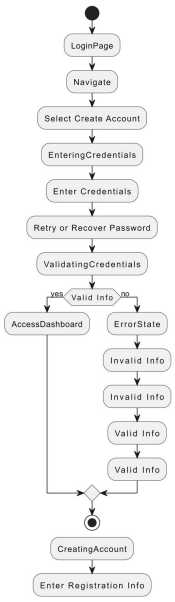


Figure 3.3: UC-1:User authentication and management state diagram

3.2 Use case 2: Search and Filtering

Related Requirements:

Efficient search and filtering of events Support for multiple filters.

Initiating Actor:

User (Adults, Fun Players, Parents for kids)

Actor's Goal:
To find events that match specific criteria.

Participating Actors:
System
Preconditions:
The user is logged in or browsing as a guest.

Post-conditions:
The user can view events matching the search/filter criteria.

Constraints:
-Results must display within 2 seconds for smooth experience.
-Support multiple filters and process complex queries efficiently.

Flow of Events for Main Success Scenario:

1. The user accesses the search/filter interface on the activity page.
2. The user enters keywords or selects filters (date, location, activity type) to refine results.
3. The system processes the query and retrieves matching events.

Flow of Events for Extensions (Alternate scenarios):

a1. No events match the criteria; the system displays a 'No results found' message and suggests other events

Use case diagram:

Use case state diagram:

Use case sequence diagram:

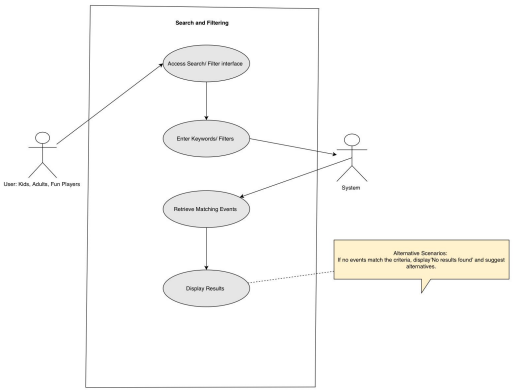


Figure 3.4: UC-2:Search and filtering

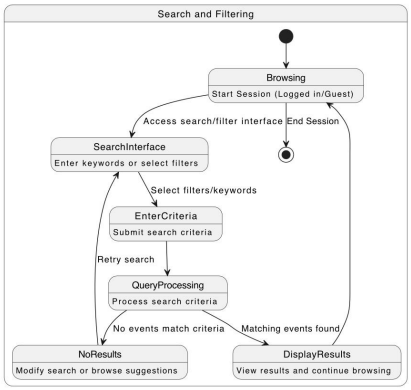


Figure 3.5: UC-2:Search and filtering state diagram

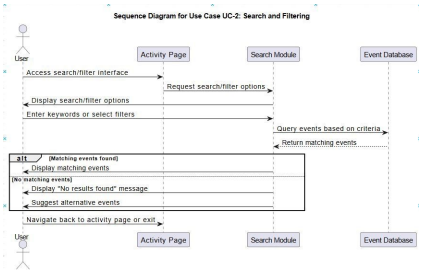


Figure 3.6: UC-2:Search and filtering sequence diagram

3.3 Use case 3: Booking and purchasing

Related Requirements:

-The system must allow guest bookings.

-Real-time synchronization of ticket availability.
-Provide secure payment options.
-Support age-appropriate content filters for kids.
-Display a confirmation page post-booking with essential details.

Initiating Actor:
User (Adults, Fun Players, Parents for kids)

Actor's Goal:
Successfully book and purchase tickets for the desired event/activity without unnecessary steps or delays.

Participating Actors:
Third parties(payment methods)
Preconditions:
The event has available tickets, and the user is logged in or can book as a guest.

Post-conditions:
User successfully books a ticket for the event.

Constraints:
-Real-Time Sync: The booking system must sync with real-time ticket availability.
-Age-Appropriate Content: Parents booking for children should have easy access to content and activities appropriate for their child's age.
-Simplified Booking Flow: The booking flow should be simple and avoid unnecessary steps to enhance user experience.

Exceptions:

-System Downtime: If the system is down during the booking, an error message is displayed asking the user to try later.
-Duplicate Booking: If a user tries to rebook the same event, the system alerts them of the existing booking

Use case diagram:

Use case state diagram:

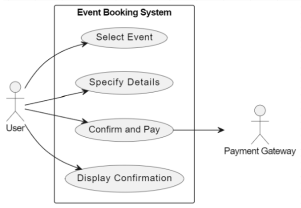


Figure 3.7: UC-3:Booking and purchasing

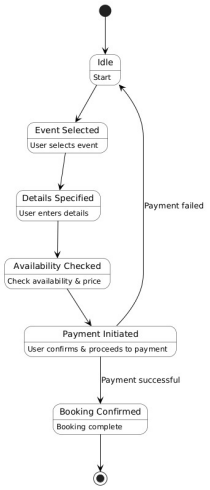


Figure 3.8: UC-3:Booking and purchasing state diagram

Use case sequence diagram:

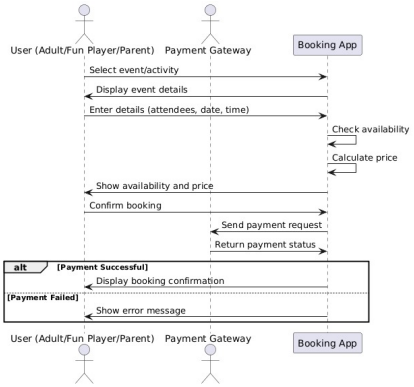


Figure 3.9: UC-3:Booking and purchasing sequence diagram

3.4 Use case 4: Event details

Related Requirements:

-The system must ensure event details are updated in real-time.
-The interface must be accessible and user-friendly across devices.

Initiating Actor:
User (Adults, Fun Players, Parents for kids)

Actor's Goal:
Successfully view the details of the selected event.

Participating Actors:
Event providers
Preconditions:
The user is logged in or browsing as a guest.

Post-conditions:
User successfully views the details of the selected event.

Constraints:
-The system must ensure event details are updated in real-time.
-The interface must be accessible and user-friendly across devices.

Flow of Events for Main Success Scenario:

1. User navigates to the events page.
2. User selects an event from the list.
3. System retrieves and displays event details, including:
- Event name
- Date and time
- Location
- Description
- Pricing
- Availability
4. User can choose to register for the event or return to the events list

Flow of Events for Extensions (Alternate scenarios):

A3a. If the event is unavailable, the system shows a message indicating this

and suggests similar events.

Exceptions:

If the system is down, an error message is displayed asking the user to try again later.
If the user tries to select a previously viewed event, the system prompts them with the latest details.

Use case diagram:

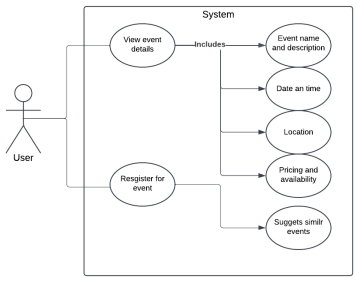


Figure 3.10: UC-4:Event details

Use case state diagram:

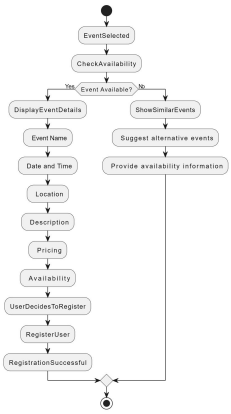


Figure 3.11: UC-4:Event details state diagram

Use case sequence diagram:

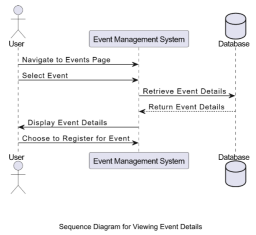


Figure 3.12: UC-2:Event details sequence diagram for viewing event details

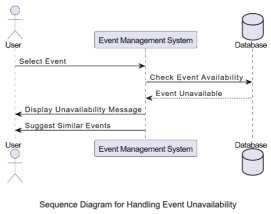


Figure 3.13: UC-2:Event details sequence diagram for handling event unavailability

3.5 Use case 5: Admin management

Related Requirements:

-The admin panel must be accessible 24/7.
-Admin should have access to detailed information regarding the events, their availability, timing, pricing, and their total sales.
-Reports should be easy to upload and export.

Initiating Actor:
Admin users.

Actor's Goal:
-To manage, initiate, edit, delete, and view events.

-To manage users.

Participating Actors:
Event organizers
Customer support agents
Third parties

Preconditions:
-Admin must be logged as an admin user with administrative privileges.
-The dashboard must be displayed with options to manage events, users, bookings, and payments.

Post-conditions:
-Events are created, deleted, edited, and updated successfully.
-User accounts and bookings are managed.
-The changes made by the admin user are reflected in the system in real time.

Constraints:
-The admin must have the proper role assigned to access sensitive information for security
-The system must ensure data integrity and be secure enough to prevent unauthorized access
-Changes made by an admin user must happen in the system in real time

Flow of Events for Main Success Scenario:

1. Admin logs into the system securely.
2. Admin is presented with an admin dashboard that allows management of events, users, bookings, payments, feedback, and reports.
3. Admin can select to manage evens and update, create, or delete them Admin can verify users, or solve a user's problem.
4. Admin can view reports and feedback to better improve the system Admin can view, cancel, and refund bookings.
5. Admin logs out and ends session

Exceptions:

-User does not have admin privileges.

Use case diagram:

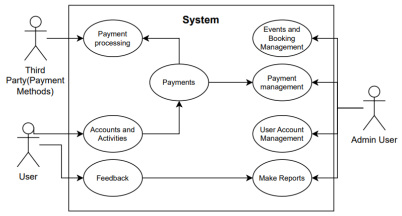


Figure 3.14: UC-5: Admin management

Use case state diagram:

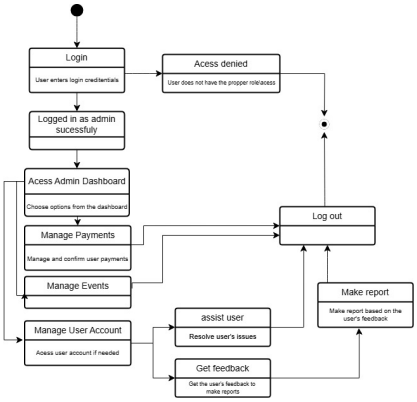


Figure 3.15: UC-5:Admin management state diagram

Use case sequence diagram:

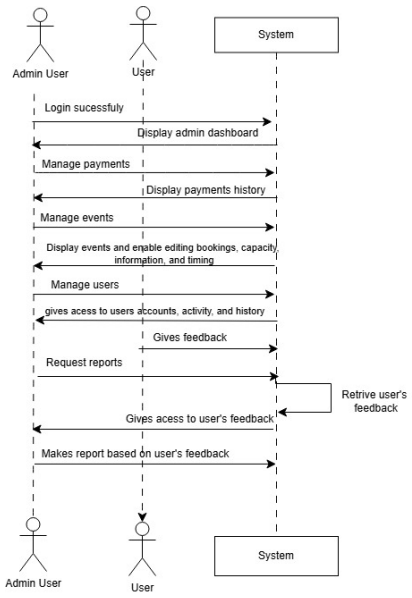


Figure 3.16: UC-5:Admin management sequence diagram

Chapter 4

Gantt Chart

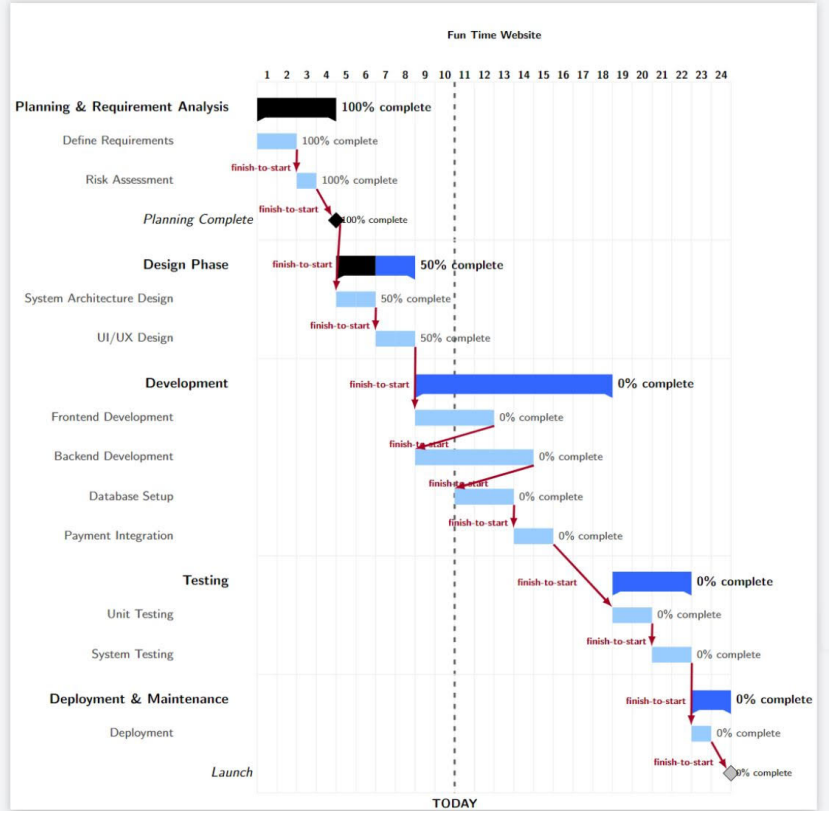


Figure 4.1: Gantt Chart

Chapter 5

UIUX

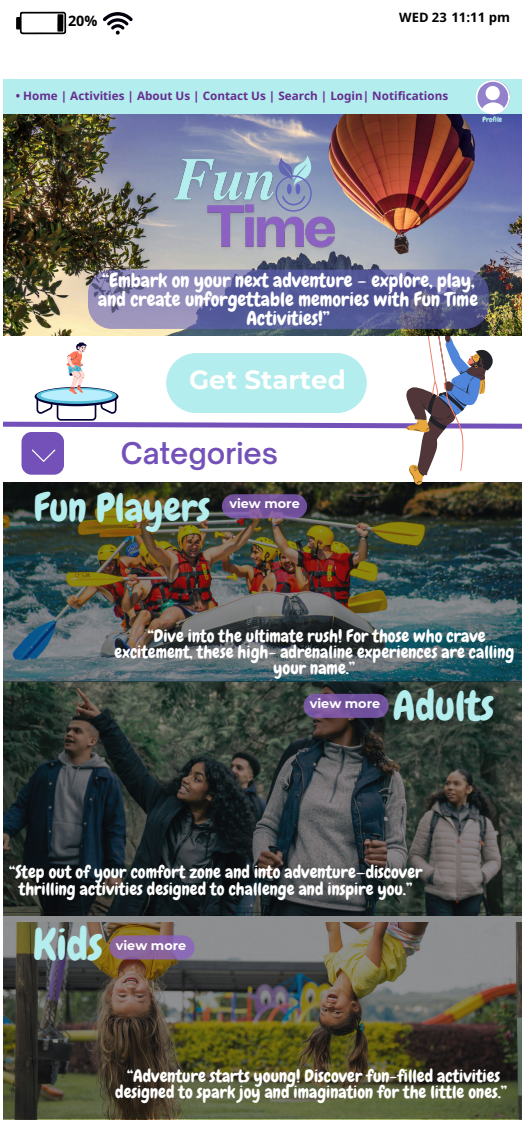


Figure 5.1: Home screen

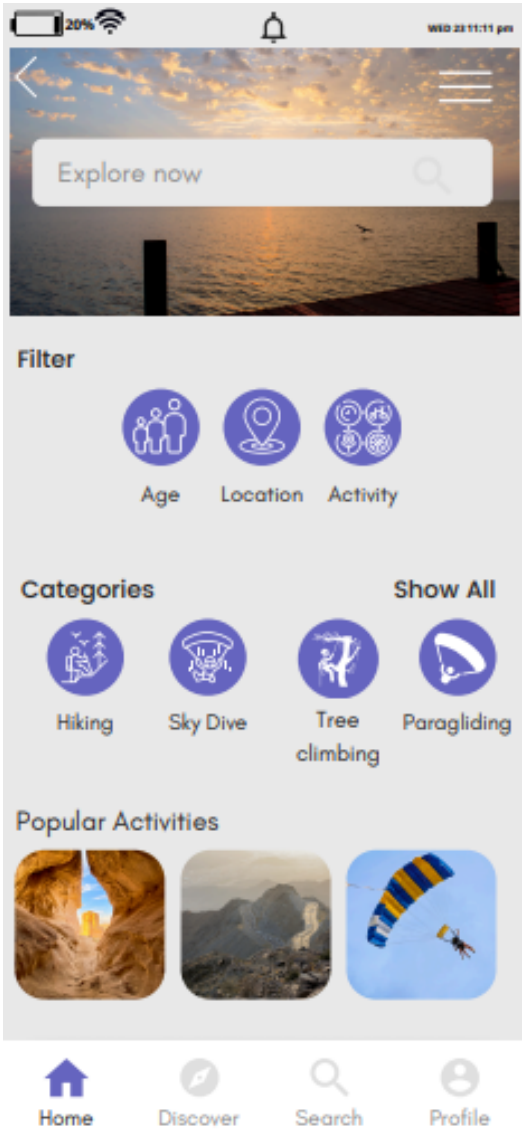


Figure 5.2: Search screen

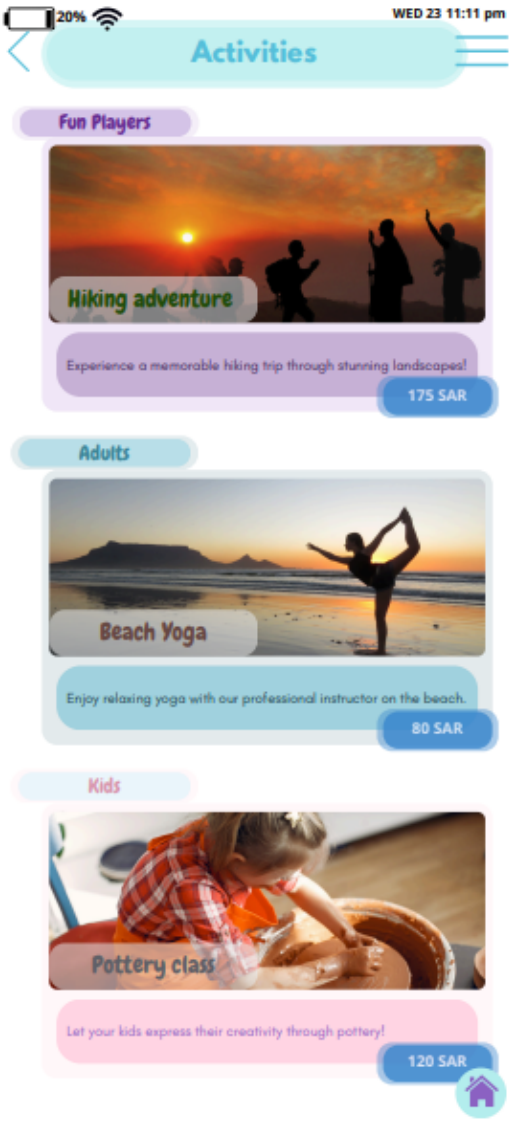


Figure 5.3: Events selection



Figure 5.4: Event details

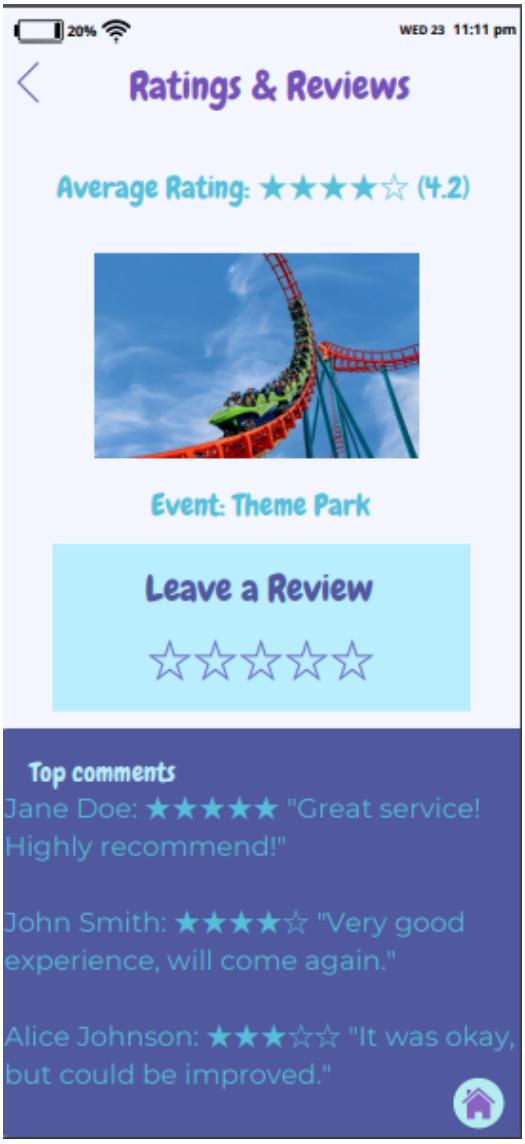


Figure 5.5: Event Reviews

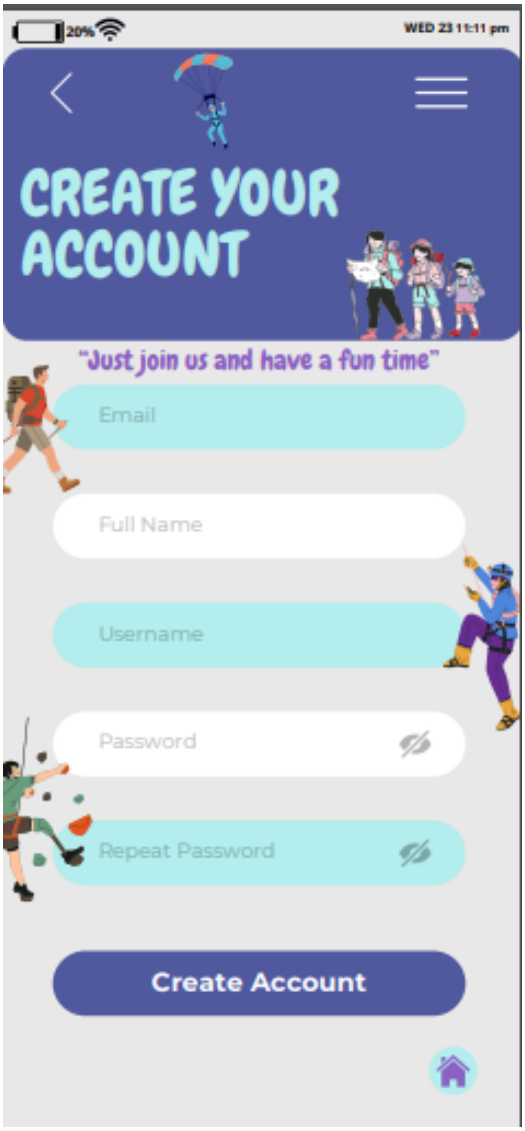


Figure 5.6: Create account

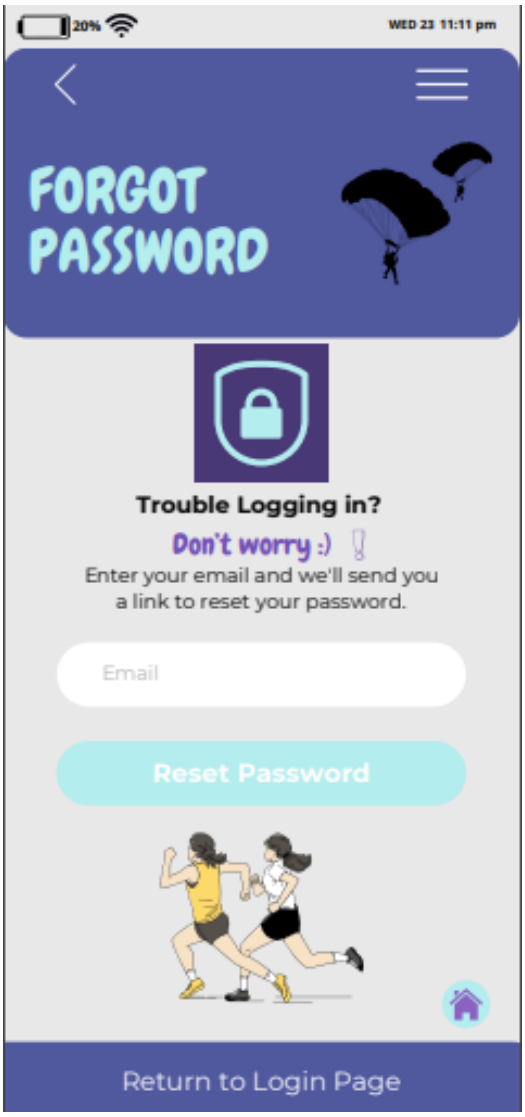


Figure 5.7: Password recovery

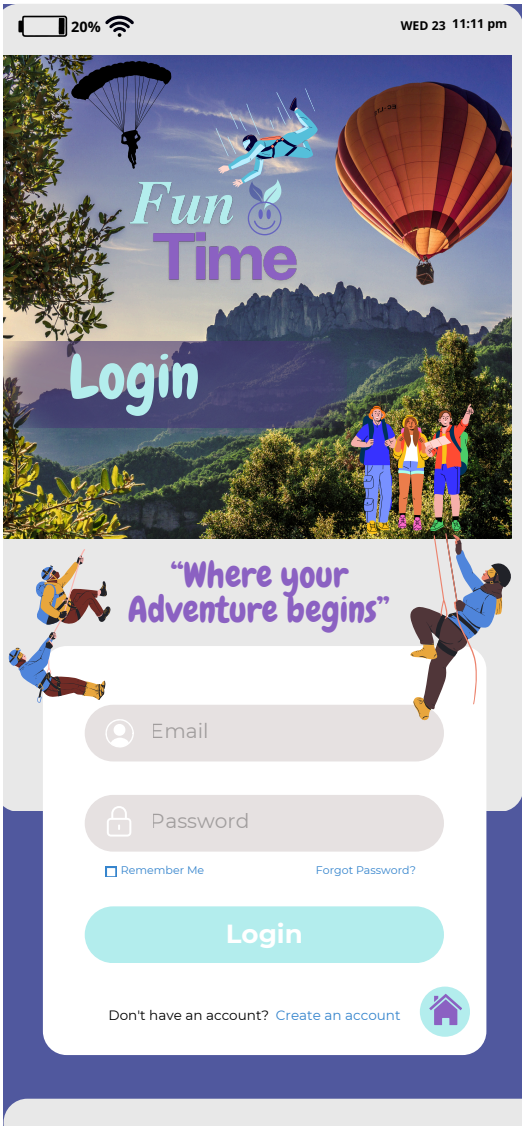


Figure 5.8: User Login



Figure 5.9: User account

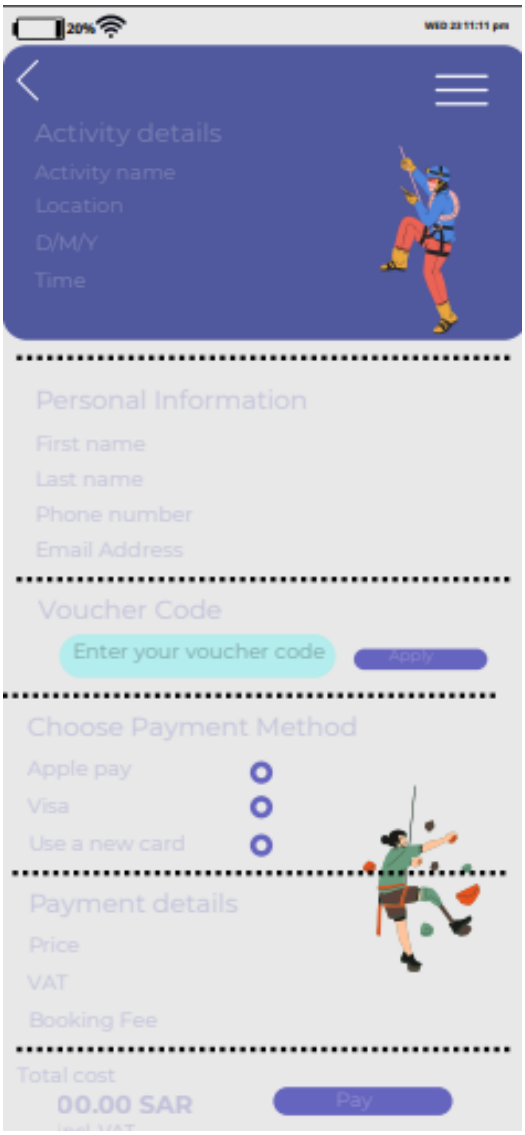


Figure 5.10: Payment processing

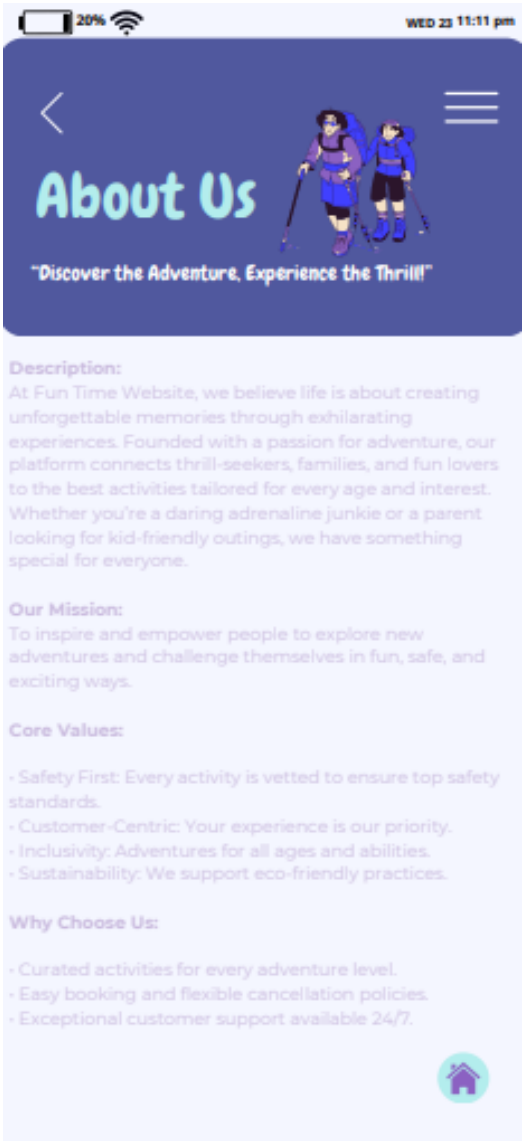


Figure 5.11: About page

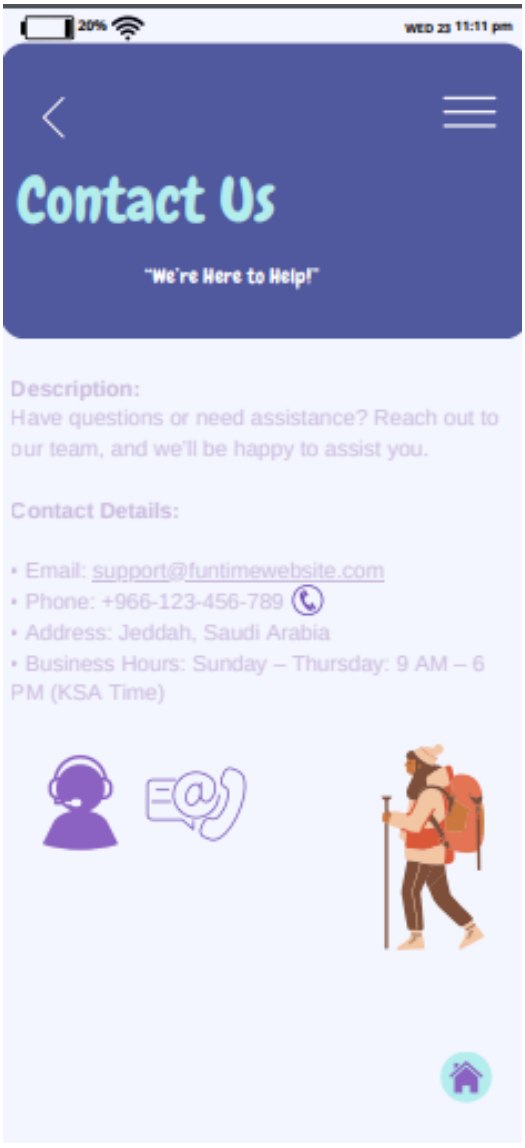


Figure 5.12: Contact information

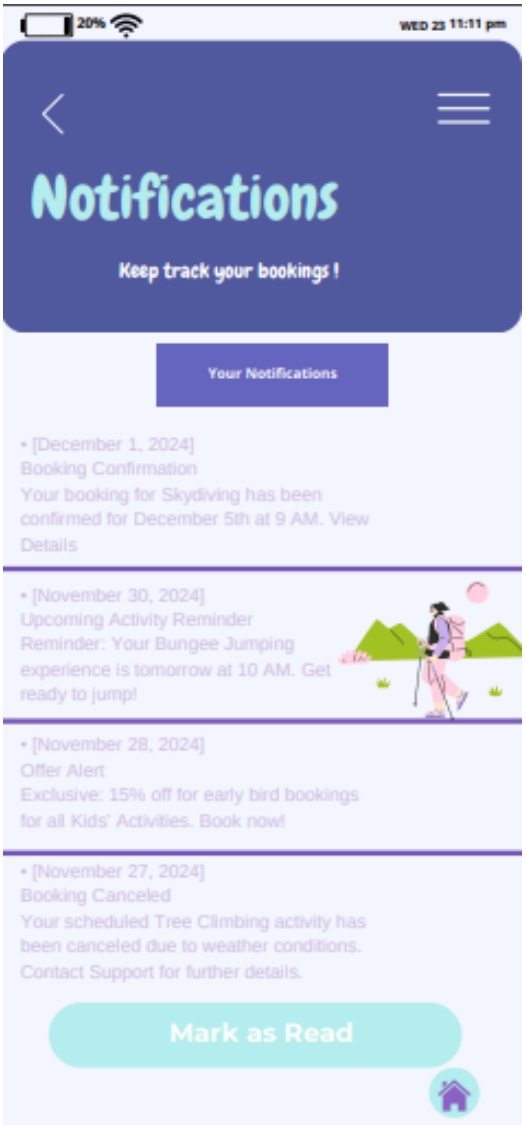


Figure 5.13: Notifications



Figure 5.14: Admin dashboard